

# Measuring Neural-Network Robustness Under Adversarial Perturbation

Experimental Evaluation of Model Behavior Under Gradient-Based Attacks

github.com/poojakira/adversarial-ml-lab - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

112

Tests passed

32.22%

Statement coverage

1,117,354

SmallCNN parameters

71.82%

Clean CIFAR-10 accuracy

SYSTEM VIEW

## Architecture and Threat-Control Flow

### Trust Boundary

#### Input surface

Data, models, requests, policies, or artifacts entering the tested path.

#### Control surface

Repository-specific validation, analysis, and fail-closed checks.

#### Evidence surface

Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01

Train/evaluate a compact CIFAR-10 classifier under a fixed seed and CPU budget.

02

Measure clean accuracy on the full CIFAR-10 test set.

03

Generate epsilon-bounded FGSM and PGD adversarial samples plus C&W L2 samples.

04

Measure robust accuracy on the committed attack subset.

05

Emit structured evidence and CI robustness-gate logic.

### Research Contribution

A reproducible robustness-evaluation harness for FGSM, PGD, and C&W that shows a measured 71.82% clean CIFAR-10 SmallCNN collapsing to 0.00% robust accuracy under PGD-20 at epsilon 8/255, while explicitly separating measured results from literature projections.

### Verified Evidence

EV-01 FGSM robust accuracy @ 8/255: 3.32%

EV-02 PGD-20 robust accuracy @ 8/255: 0.00%

EV-03 C&W L2 robust accuracy: 4.20%

EV-04 Robust attack evaluation subset: 1,024 samples

### Method and Control Path

01 Train/evaluate a compact CIFAR-10 classifier under a fixed seed and CPU budget.

02 Measure clean accuracy on the full CIFAR-10 test set.

03 Generate epsilon-bounded FGSM and PGD adversarial samples plus C&W L2 samples.

04 Measure robust accuracy on the committed attack subset.

05 Emit structured evidence and CI robustness-gate logic.

### Limitations and Residual Risk

- This repository measures vulnerability; it does not make a model robust.
- Robust-accuracy attacks use a 1,024-sample subset, not the full 10,000 test set.
- No AutoAttack, black-box transfer, or certified-robustness result is established.
- results/cifar10\_resnet18\_benchmark.json is explicitly literature-projected/synthetic and must not be presented as measured.

### Reproducibility

Expected evidence at the cited snapshot: 109 passed, 32.22% coverage.

### Evidence Provenance

#### Claim discipline

Metrics come from the repository's verified metrics and research brief.

#### Scope discipline

Benchmarks are reported only at their documented fixture, dataset, or test scope.

#### Reproduction

PDF and PNG are generated locally from committed evidence without paid services.